Should we discount tomorrow?

Forecasting next-day sales to help a Dingdong category manager decide **whether to discount and by how much.**

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01 Start with a decision the data can support

Fresh food loses value quickly. Too little discounting can leave stock unsold; too much gives up sales value. We study daily discount choices using public retail histories.

FreshRetailNet-50K

Sales, discounts, stockout hours and calendar information match the inputs this decision needs.

50,000 store-product histories → **16,684** eligible histories → **5** selected stores → **312** selected histories

Eligibility: 15-85% discounted days (rate < 0.95); 6+ discount-state changes; fewer than 20% zero-sales days.
Choose the five stores with the most eligible histories. One city; cohort selected using the full 90-day training file.

02 Predict sales first. Then choose a price.

PAST + PLANNED INPUTS

Recent sales and stockouts, past discounts and activity, tomorrow's calendar and **candidate price rate.**

→

TWO LEARNERS, ONE FORECAST

Random Forest averages trees fit to bootstrapped samples. **CatBoost** corrects errors sequentially. **Final = 0.5 RF + 0.5 CatBoost.**

→

SUPPORTED PRICE CHOICE

Compare eight price rates. Keep rates with historical support, apply stock rules, then maximize **rate x forecast.**

The target is normalized observed sales, not units or unconstrained demand. Rate 0.95 means 5% off.
Historical support: at least 3 training days within 0.025 of a rate. The default value proxy assumes zero unit cost.

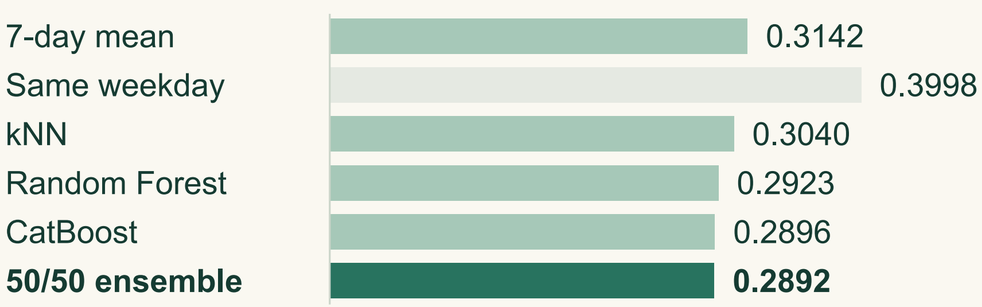
EVALUATION — 26 Jun - 2 Jul 2024

5 expanding weekly validation folds → **Refit on 25,896 rows** → **2,184 next-day cases**

03 A better sales forecast

7.9% lower final-week MAE than the seven-day mean

Mean absolute error (MAE); lower is better



Root mean square error (RMSE): 0.4593

Weight experiment: RF shares of 40-70% give validation MAE 0.30118-0.30144. We keep 50/50 for simplicity; it adds little over CatBoost alone.

04 Pricing still needs rules

Maximizing predicted value suggests a discount almost every day. Assumed stock rules change that behaviour.

Share of cases recommending a discount



Stockout yesterday: keep full price.
No stockout in 7 days: require a supported discount.
Otherwise: let the model compare eligible rates.

Same 2,184 cases. These are policy assumptions, not evidence of inventory age or higher revenue.

05 ONE PRODUCT, ONE DAY

5% off

recommended discount

Store 18 / Product 11 / 26 June 2024

Yesterday: no stockout. Two stockout days in the past week.
The model compares five historically supported rates.

Full price **1.12** → 5% off **1.26** predicted sales

Illustrative model scenario; not observed sales uplift.

Value proxy
0.95 x 1.26 = 1.20
vs. 1.12 at full price

A decision aid, with business impact still to test.

Discounts are observational and stockouts can hide demand. We have no stock-age or waste outcomes; the final week was reused during development. Next: actual inventory/cost data and a controlled store pilot.

Data: FreshRetailNet-50K / Dingdong / CC BY 4.0

Final notebook: pricing-ml-ap @ a7e26dd | Code + Docker

EXPLORE THE PROJECT